BIOCHEMISTRY MIDTERM EXAMINATION

Course Code: BCH-001 | Duration: 60 minutes | Total Marks: 50

Student Name: _____ **Roll No.:** _____

Class / Section: _____ **Date:** _____

Instructions

1. Attempt all questions.
2. For Section A, select **one** best answer for each question by circling or clearly marking A, B, C or D.
3. For Sections B and C, write your answers in the spaces provided. Keep your responses clear and relevant.
4. The marks for each question are shown beside the question. Manage your time accordingly.

Section A — Multiple-Choice Questions (12 × 1 = 12 marks)

Choose the one best answer for each question.

Q1. Which element forms the backbone of most organic molecules in living organisms? (1 mark)

A) Carbon

B) Nitrogen

C) Sulfur

D) Phosphorus

Q2. Which statement best describes the scope of biochemistry? (1 mark)

A) The study of how organisms interact with their ecosystems

B) The study of the physical forces acting on living tissues

C) The study of the chemical substances and reactions found in living organisms

D) The study of classifying organisms by evolutionary ancestry

Q3. The everyday term 'sugars' most commonly refers to which classes of carbohydrate? (1 mark)

A) Polysaccharides only

B) Monosaccharides and disaccharides

C) Amino acids and peptides

D) Fatty acids and glycerol

Q4. A patient's intestinal sample is shown to break sucrose into two monosaccharides. Which enzyme is responsible, and which products are formed? (1 mark)

A) Lactase; glucose and galactose

B) Maltase; two glucose molecules

C) Amylase; smaller sugars and ultimately glucose

D) Sucrase; glucose and fructose

Q5. Why is the highly branched structure of glycogen biologically advantageous? (1 mark)

A) It makes glycogen insoluble so it cannot leave the cell

B) It allows glycogen to serve as a structural support material

C) It allows glucose to be mobilised quickly

D) It enables humans to digest cellulose using their own enzymes

Q6. Cellulose passes through the human digestive system largely undigested. Which explanation from the material best accounts for this? (1 mark)

A) It has beta-1,4 linkages, which humans cannot break using their own enzymes

B) It is a highly branched molecule that resists enzyme action

C) It is a protein rather than a carbohydrate, so no carbohydrase acts on it

D) It dissolves readily in water and is absorbed intact without digestion

Q7. Which hormone lowers blood glucose by promoting its uptake and storage in responsive tissues? (1 mark)

A) Glucagon, which stimulates the liver to release glucose

B) Insulin, which promotes uptake and storage of glucose in responsive tissues

C) Amylase, which breaks starch into smaller sugars

D) Pepsin, which acts on proteins under acidic conditions

Q8. Which set of elements is found in proteins? (1 mark)

A) Carbon, hydrogen and oxygen only

B) Carbon, hydrogen, oxygen and phosphorus

C) Carbon and oxygen only

D) Carbon, hydrogen, oxygen and nitrogen, and in some cases sulfur

Q9. Which statement best distinguishes denaturation from hydrolysis of a protein? (1 mark)

A) Both denaturation and hydrolysis break covalent peptide bonds using water

B) Denaturation breaks the primary amino-acid sequence, whereas hydrolysis only unfolds the protein

C) Denaturation changes the protein's folded shape and usually leaves the primary amino-acid sequence intact, whereas hydrolysis breaks covalent peptide bonds using water

D) Denaturation is always reversible and hydrolysis is always irreversible, regardless of conditions

Q10. An enzyme solution is heated gently and its activity falls sharply, yet no new amino acids appear in the solution. Which explanation best fits these observations? (1 mark)

A) The enzyme has been hydrolysed completely into individual amino acids

B) The enzyme's peptide bonds have been cleaved by water

C) The enzyme has been digested by pepsin under acidic conditions

D) The enzyme has been denatured, so its folded shape and therefore its activity are altered

Q11. In RNA, which nitrogen-containing base takes the place of the thymine found in DNA? (1 mark)

A) Uracil

B) Guanine

C) Cytosine

D) Adenine

Q12. A nucleic acid sample is found to contain ribose and uracil. Which conclusion is best supported by this evidence? (1 mark)

A) It is DNA storing long-term genetic information

B) It is RNA, which functions in gene expression and other cellular functions

C) It is a double-stranded helix containing A, T, G and C

D) It is a protein built from nucleotides joined by peptide bonds

Section B — Short-Answer Questions (8 × 2.5 = 20 marks)

Answer each question in the space provided.

Q13. Define biochemistry and explain why a knowledge of it is important in medicine and in everyday life. (2.5 marks)

Q14. State what is meant by a sugar, name the two classes the term commonly refers to, and give an example of each with the elements they contain. (2.5 marks)

Q15. Compare starch, glycogen and cellulose with respect to structure, digestibility in humans and biological role. (2.5 marks)

Q16. List the main functions of proteins in the body and state the elements commonly found in them. (2.5 marks)

Q17. Compare DNA and RNA in terms of their sugar, nitrogenous bases, typical structure and main role. (2.5 marks)

Q18. Describe the main stages of protein digestion, naming the enzymes and the location of each step. (2.5 marks)

Q19. Distinguish between denaturation and hydrolysis of a protein, stating what each process changes and whether each is reversible. (2.5 marks)

Q20. Evaluate the statement: 'Blood glucose is controlled mainly by insulin and glucagon.' Justify your answer using the roles of these hormones and the fate of glucose in cells. (2.5 marks)

Section C — Long-Answer Questions (2 × 9 = 18 marks)

Answer both questions. Organise your responses into clear, well-developed paragraphs. You may use labelled diagrams or tables where appropriate.

Q21. Compare starch, glycogen and cellulose in terms of their structure, glycosidic linkages and biological roles. Analyse why humans can break down starch and glycogen but not cellulose, and justify the nutritional consequences of this difference. (9 marks)

Q22. Distinguish between the denaturation and the hydrolysis of proteins with reference to protein structure and bonding. Evaluate why this distinction is essential for explaining both the loss of enzyme activity and the digestion of dietary protein. (9 marks)

End of Examination — Please check that you have attempted all questions.